# Vishwanath Vinod

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## Education

**Indian Institute of Technology Madras** — *2022 - Present*
**Dual Degree (B.Tech in Electrical Engineering + M.Tech in Data Science & Artificial Intelligence)** — ***9.74/10.0***

**Kuriakose Elias HSS Mannanam** — *2020 - 2022*
**Class XII - Kerala HSE** — ***99.83%***

## Publications

**Point Neuron Embedded Kalman Filter for Narrowband Sound Source Tracking** — *EUSIPCO 2026*
**Vishwanath Vinod**, S. Xu, P. Samarasinghe, A. Bastine, Thushara Abhayapala

**Cooperative Multi-Agent Reinforcement Learning for Fair Dynamic Pricing** — *Computing Conference 2026*
**Vishwanath Vinod**, Rachel Kalpana Kalaimani

## Awards

**Future Research Talent Award** — *2025*
**Awarded by the Australian National University during my research internship**

**Research Spotlight Certificate of Excellence** — *2024 - 2025*
**Awarded for the best undergraduate research projects in the department**

**Sri V. Rajagopalan Memorial Award** — *2023 - 2024*
**Awarded for the highest academic performance in the department during sophomore year**

## Research Experience

### Privatizing State Space Models — *Mar '26 - Present*

**Masters Project Under Prof. Krishna Pillutla**

- Conducted an extensive literature review on State Space Models (SSMs) and delivered a series of **research talks** on the topic.
- Implemented **MAMBA from scratch** in PyTorch, including the parallelized selective scan algorithm.
- Incorporated **Privacy amplification by iteration** into the contractive steps of SSMs and derived privacy guarantees for a single-layer.
- Characterized privacy leakage and extended it to **multi-layer SSMs**, including models with input-dependent parameters like MAMBA.
- Empirically validated the synthetic text generation performance of this private model by evaluating it on the **WikiText datasets**.

### Point Neuron Embedded Kalman Filter (PNEKF) — *May '25 - Sep '25*

**Research Intern, Australian National University under Prof. Thushara Abhayapala**

- Implemented Point Neuron Learning in PyTorch and extended it to **3-D sound field reconstruction** in reverberant settings.
- Designed a Point Neuron–based **Orthogonal Matching Pursuit (OMP)** algorithm for sound source localization to initiate tracking.
- Proposed a **novel Kalman Filter variant** that integrates the point neuron framework to facilitate **multi-source tracking**.
- Simulated reverberant room acoustics using the **Image Source Model (ISM)** pipeline in MATLAB for empirical evaluation.
- **Experimentally validated** PNEKF's superiority over baseline methods across diverse frequencies, sources, and microphone settings.

### Cooperative Multi-Agent Reinforcement Learning for Fair Dynamic Pricing — *Apr '24 - Nov '24*

**Undergraduate Research Project under Prof. Rachel Kalpana Kalaimani**

- Designed a custom OpenAI Gym environment for dynamic pricing with **finite inventory** and a **limited selling horizon**.
- Conducted a comprehensive literature review on dynamic pricing, fairness metrics, and MARL methods (**IRAT**, **Dec-MADDPG**).
- Developed a **cooperative MARL** framework with a **dual-reward** formulation to promote **group fairness** in dynamic pricing.
- Introduced a novel **control parameter** to tune the fairness–profit trade-off in this multi-objective optimization setting.
- **Experimentally validated** the proposed algorithm's superiority through a comprehensive analysis against the existing benchmark.

## Relevant Coursework

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| Machine Learning | Introduction to Reinforcement Learning, Privacy in AI, Advanced Optimization, Natural Language Processing<br>Learning from Human Preferences, Deep Learning for Imaging, Introduction to Machine Learning,<br>Applied Programming Lab, Deep Learning Specialization, Data Analytics Laboratory |
| Electrical Engineering | Control Systems & Engineering, Signals & Systems, Digital Signal Processing, Image Signal Processing<br>Communication Systems, Analog Systems & Lab, Electrical Machines & Lab, Microprocessor & Lab |
| Mathematics | Applied Linear Algebra for Electrical Engineering, Probability Foundations for Electrical Engineers,<br>Statistics for Electrical Engineers, Computational Physics, Differential Equations |
| Computer Science | Object Oriented Programming Specialization, Data Structures & Algorithms, Computer Organization |

## Teaching Assistant

**DA2401: Foundations of Machine Learning (Prof. Krishna Pillutla)** *Jul '26 - Present*
**DA6400: Introduction to Reinforcement Learning (Prof. Balaraman Ravindran)** *Jan '26 - May '26*

## Scholastic Achievements

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| Opening Rank | Enrolled in the Electrical Engineering department at IIT Madras as the department's **highest-ranked** admit. |
| JEE Advanced 2022 | Secured **All India Rank 252** among 200,000 candidates, placing in the top **0.2 %tile** of candidates |
| JEE Main 2022 | Secured **All India Rank 506** among 1.1 million candidates, with a percentile score of **99.95 %tile** |
| KEAM 2022 | Secured **State Rank 1** in the Kerala Engineering Entrance Exam among 73,000 candidates |
| KVPY 2020 | Secured **All India Rank 30** (SA stream); received the KVPY Fellowship from the Government of India |
| Olympiads | Qualified for the **Indian National Mathematics Olympiad** 2019 and the **National Junior Science Olympiad** 2018. |

## Major Course Projects

### Machine Leaning Techniques *Jul '25 - Nov '25*

**DA5401 - Data Analytics Lab under Prof. Sudarshan Santhiappan**

- Mitigated class imbalance via **Clustering (CBO, CBU)**, **SMOTE**, and **GMM-based synthetic resampling**.
- Analyzed multi-label manifold structures and data veracity using **t-SNE** and **Isomap** embeddings.
- Implemented data imputation strategies including **Linear/Non-Linear Regression**, Median, and Listwise deletion.
- Evaluated multi-class model performance using macro/micro **ROC** and **Precision-Recall** curves.
- Built **similarity regression pipelines** for AI prompt-responses under extreme class imbalance and limited sample constraints.
- Boosted accuracy using rank normalization, metric shuffling, dimensionality reduction, and ensemble modeling techniques.

### Image Denoising & Privacy in Lensless Imaging *Jul '25 - Nov '25*

**EE5175 - Image Signal Processing & EE5179 - Deep Learning from Imaging under Prof. A.N. Rajagopalan**

- Implemented **BM3D** to denoise images from the **RENOIR dataset** and low-light smartphone video frames.
- Benchmarked BM3D against classical filters (NLM, Bilateral, Gaussian) using **PSNR** and **SSIM** metrics.
- Implemented classical image processing tasks including segmentation, mosaicing, occlusion detection, deblurring and depth mapping.
- Surveyed lensless imaging architectures (**FlatCam**, **PhlatCam**, **FlatNet**, **OpEnCam**, **CAnOpIC**) and face recognition models.
- Conducted ablation studies of these models (**DeepFace, ArcFace**) on LFW dataset under varying noise levels in **Wiener deconvolution**.

### Reinforcement Learning Methods *Jan '25 – May '25*

**DA6400 - Introduction to Reinforcement Learning under Prof. Balaraman Ravindran**

- Conducted experimental studies of multiple RL algorithms across diverse use cases.
- Analyzed the performance of **SARSA** and **Q-Learning** on a modified grid-world under varied configurations.
- Implemented **Dueling-DQN**, **MC-Reinforce**, and **Actor-Critic** methods on OpenAI Gymnasium tasks (*CartPole, AcroBot, MountainCar*).
- Performed comparative evaluation of hierarchical RL algorithms such as **SMDP** and **Intra-option Q-Learning** on the Taxi environment.

### Information Retrieval System *Jan '25 – May '25*

**CS6370 - Natural Language Processing under Prof. Sutanu Chakraborty**

- Implemented and evaluated classical (**TF–IDF**, **BM25**), semantic (**LSA**, **ESA**, **NESA**), and hybrid IR models on the Cranfield dataset.
- Incorporated semantic enhancements such as **WordNet-based query expansion** and dense transformer embeddings (**All-MiniLM**).
- Designed a modular preprocessing pipeline with sentence segmentation, tokenization, stemming, and stopword removal.
- Built **query autocomplete** using N-gram and Trie models with a top-$k$ interactive interface robust to typos and incomplete input.
- Conducted comparative evaluation with **MAP@10**, **Precision@10**, **Recall@10**, **nDCG@10**, and significance tests on query-level scores.

## Talks & Presentations

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| 2026 | State Space Models Research Talk @ Machine Learning & AI Lab, IIT Madras |
| 2026 | Oral Presentation at the 14th Computing Conference @ London, UK |
| 2026 | Speaker at the Electrical Engineering Research Symposium @ IIT Madras |
| 2025 | Research Presentation @ Audio & Acoustic Signal Processing Group, Australian National University |

## Co-Curriculars

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| 2025-26 | Reviewer @ Indian Control Conference 2026 |
| 2024–25 | Coordinator @ Electrical Engineering Research Club, IIT Madras |
| 2023–24 | Deputy Placement Coordinator @ Placement & Internship Cell, IIT Madras |
| 2022–24 | Coordinator @ Coding & Logic, Shaastra Tech Fest 2024, IIT Madras |